

Agenda

1. Friday Feedback
2. Announcements
 - Exam 1 Scores Available
 - Regrade Requests
 - First Year Student Midterm Grades
3. Section 11.2



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Series

We will now investigate summing terms together as an
infinite series: $a_1 + a_2 + a_3 + \dots = \sum_{n=1}^{\infty} a_n$

Define the n^{th} partial sum as $S_n = a_1 + a_2 + \dots + a_n$
 $S_n = \sum_{i=1}^n a_i$

We will focus on the limit $\lim_{n \rightarrow \infty} S_n$

if it converges/diverges, the infinite series will also.

Activity: Write each infinite series using summation notation

1.) $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots = \sum_{n=1}^{\infty} \frac{1}{n}$

2.) $\frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \frac{2}{81} + \dots = \sum_{n=1}^{\infty} 2\left(\frac{1}{3}\right)^n$

3.) $0 + 1 + 0 + 1 + \dots = \sum_{n=1}^{\infty} \left[\frac{1}{2} + \frac{1}{2}(-1)^n \right]$

Definitions Let $S_n = a_1 + a_2 + a_3 + \dots + a_n$ be the n^{th} partial sum

1.) We say $\sum_{n=1}^{\infty} a_n$ converges if $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i$ converges as a sequence.

2.) We say $\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i$ diverges as a sequence.

Examples: Determine whether each series Converges or diverges

1.) $\sum_{n=1}^{\infty} (-1)^n$ diverges $\rightarrow \{S_n\} = \{-1, 0, -1, 0, \dots\}$
 $\therefore$ series diverges

2.) $\sum_{n=1}^{\infty} 0$ converge $\{S_n\} = \{0, 0, 0, \dots\}$
 $\therefore$ series converges

3.) $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \frac{1}{2^n}$ $S_1 = \frac{1}{2}$ $S_2 = \frac{3}{4}$ $S_3 = \frac{7}{8}$
 $S_n = \frac{2^n - 1}{2^n}$

$$\lim_{n \rightarrow \infty} S_n = 1$$

converges to 1

Important Series

Geometric Series Let a, r be constants where $a \neq 0$;
We call a series of the form $\sum_{n=1}^{\infty} ar^{n-1}$ a geometric series
 $\rightarrow a + ar + ar^2 + \dots$

We can find a formula for the n^{th} partial sum of a geometric series:

$$\begin{aligned} S_n &= a + ar + ar^2 + \dots + ar^{n-2} + ar^{n-1} \\ - \quad (rS_n &= ar + ar^2 + ar^3 + \dots + ar^{n-1} + ar^n) \\ \hline S_n - rS_n &= a - ar^n \end{aligned}$$

$$\begin{aligned} S_n(1-r) &= a(1-r^n) \\ S_n &= a \left(\frac{1-r^n}{1-r} \right) \end{aligned}$$

We can use $\lim_{n \rightarrow \infty} S_n$ to determine when $\sum_{n=1}^{\infty} ar^{n-1}$

Converges or diverges.

1.) If $-1 < r < 1$, then $\lim_{n \rightarrow \infty} r^n = 0$, so $\lim_{n \rightarrow \infty} a \left(\frac{1-r^n}{1-r} \right) = \frac{a}{1-r}$
($|r| < 1$)

So if $-1 < r < 1$, then $\sum_{n=1}^{\infty} ar^{n-1}$ converges to $\frac{a}{1-r}$

2.) If $r > 1$ or $r \leq -1$, then $\lim_{n \rightarrow \infty} r^n$ does not exist,
($|r| \geq 1$)

So $\lim_{n \rightarrow \infty} a \left(\frac{1-r^n}{1-r} \right)$ DNE

and if $|r| \geq 1$, then $\sum_{n=1}^{\infty} ar^{n-1}$ diverges.

Example: Determine whether the series converges or diverges.

$$1.) \sum_{n=1}^{\infty} 3^{n+2} 2^{n+1} \stackrel{\text{goal}}{=} \sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=1}^{\infty} 4(2/3)^{n-1}$$

$$a_1 = 3^1 2^2 = 12$$

$$a_3 = 16$$

$$2.) \sum_{n=0}^{\infty} \frac{4^{2n}}{(-5)^{n-1}}$$

$$a_2 = \frac{3^0 (2^3) = 8}{ar}$$

$$8 = 12r$$

$$2/3 = r$$

$$3.) \sum_{n=3}^{\infty} 7\left(\frac{1}{3}\right)^{n-1}$$

Harmonic Series: